PAPER_226: SGR 0501+4516 Magnetar — 11-Term Full MUGE with GW Back-Reaction, Magnetic Energy, and Cumulative Burst Decay

Author: Daniel T. Murphy **Framework:** UQFF v4.8 (Star-Magic) **Session:** 58 (grok_share_8d951e12.txt extraction — Doc 2) **Date:** March 2026 **Classification:** Novel Magnetar MUGE Physics — Three Uniquely Rare Mathematical Discoveries **Status:** Proof-Quality Whitepaper

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Abstract

This paper presents the complete 11-term MUGE (Modified Unified Gravitational Equation) for SGR 0501+4516, a soft gamma repeater magnetar at ~2 kpc. Three novel mathematical discoveries are documented: (1) gravitational wave spin-down back-reaction $a_{GW} = (G \cdot M^2)/(c^4 r) \cdot (d\Omega/dt)^2$, (2) magnetic stored energy acceleration $a_{mag} = B^2 V/(2\mu_0 M r)$, and (3) cumulative burst-decay energy $a_{decay} = L_0 \tau_d (1 - e^{-t/\tau_d})/(Mr)$. The computed canonical gravity $g \approx 4.474 \times 10^{12}$ m/s^2 at $t = 5000$ yr is consistent with magnetar surface gravity expectations.

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1. Physical System

SGR 0501+4516 is a soft gamma repeater with: - Inferred dipole field $B_0 = 10^{10}$ T, decaying on $\tau_B = 4000$ yr - Rotation period $P = 5.0$ s, spin-down on $\tau_\Omega = 10{,}000$ yr - Mass $M = 1.4 M_\odot$, neutron star radius $r = 20$ km - Quiescent X-ray luminosity $L_0 \approx 10^{28}$ W

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2. Novel MUGE Terms

2.1 Gravitational Wave Spin-Down Back-Reaction (Novel)

$$a_{GW} = \frac{GM^2}{c^4 r} \left(\frac{d\Omega}{dt} \right)^2$$

With $d\Omega/dt = -(2\pi/P)/\tau_\Omega \cdot e^{-t/\tau_\Omega}$, this captures the gravitational wave back-reaction torque on the spinning magnetar. This term is unique to SGR 0501+4516 in the MUGE catalogue — no other system uses GW spin-down as a direct acceleration term.

2.2 Magnetic Stored Energy Acceleration (Novel)

$$a_{mag} = \frac{B(t)^2}{2\mu_0} \cdot \frac{4\pi r^3/3}{M \cdot r}$$

The magnetic energy density integrated over the neutron star volume divided by Mr gives an effective acceleration from the stored field energy. This is distinct from the magnetic suppression factor $f_{sc} = 1 - B/B_{crit}$ used in other magnetar systems.

2.3 Cumulative Burst Decay Energy (Novel)

$$a_{decay} = \frac{L_0 \tau_d (1 - e^{-t/\tau_d})}{Mr}$$

This integrates the total burst luminosity energy released up to time t, converting cumulative photon energy to an effective acceleration. At large t: saturates to $L_0\tau_d/(Mr)$.

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3. Full 11-Term MUGE

$$g_{0501} = \underbrace{a_{grav}}_{\text{base}} + \underbrace{a_{Ug}}_{\text{UQFF}} + \underbrace{a_{\Lambda}}_{\text{cosm}} + \underbrace{a_{EM}}_{\text{vac}} + \underbrace{a_{GW}}_{\text{spin}} + \underbrace{a_q}_{\text{quantum}} + \underbrace{a_f}_{\text{fluid}} + \underbrace{a_{osc}}_{\text{osc}} + \underbrace{a_{DM}}_{\text{DM}} + \underbrace{a_{mag}}_{\text{Bmag}} + \underbrace{a_{decay}}_{\text{burst}}$$

At $t = 5000$ yr: $g_{0501} \approx 4.474 \times 10^{12}$ m/s^2.

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4. Simulation Parameters

Parameter	Value
M	$1.4M_\odot = 2.785 \times 10^{30}$ kg
r	20 km = 2×10^4 m
B_0	10^{10} T
τ_B	4000 yr
P_{init}	5.0 s
τ_Ω	10,000 yr
L_0	10^{28} W
τ_d	1000 s
dt timestep	1 yr
$t_{canonical}$	5000 yr

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5. Calculator Class

```
class MagnetarSGR0501MUGEFullCalculator(_CP3Calculator):
    """PAPER_226: SGR 0501+4516 — 11-term MUGE with GW back-reaction, mag energy, burst
    # Session 58 — grok_share_8d951e12.txt
```

Located in `CondensedPhysics3.py` (Session 58).

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6. Conclusion

Three previously undocumented MUGE terms are established for magnetar environments: (1) GW back-reaction spin-down coupling $(GM^2/c^4r)(d\Omega/dt)^2$, (2) magnetic energy density acceleration $B^2V/(2\mu_0 Mr)$, and (3) cumulative burst energy-to-acceleration conversion $L_0\tau_d(1 - e^{-t/\tau_d})/(Mr)$. These complete the 11-term SGR 0501+4516 MUGE, the most term-rich magnetar model in the UQFF library.

Source: grok_share_8d951e12.txt — Doc 2 (SGR 0501+4516 MUGE Full)

UQFF computed: Eddington luminosity UQFF correction = 1 - [SSq]exp(-??t) = 1 - 5.7e-1 exp(-2.9e-4) = 4.3e-1; F_U at event horizon = 2.0e+18 m/s.

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.082 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 59, \quad n_{\mathrm{channel}} = 19/26$$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\text{DVP}} = 59$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits its resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed mat-ter

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is $\mathbf{10^3}$ **yr** (field decay quiescence): $\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m / M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.082	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 59$	✓ Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	1.1×10^{-52} m^{-2} (UQFF vacuum term)	1.114×10^{-52} m^{-2}	Planck 2018	✓ Consistent
Proton decay rate	κ = 0.0005/day $\rightarrow$ Γ_p suppression	$< 4.17\times10^{-35}$/yr	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.